Case Study 1: Online Shopping Order System

Sample Data

```
> db.orders.insertMany([
  {
    customer_id: "C101",
    name: "Asha",
    items: [
      { product: "Mobile", price: 15000 },
      { product: "Earphones", price: 2000 }
    ]
  },
  {
    customer_id: "C102",
    name: "Kiran",
    items: [
      { product: "Mobile", price: 15000 },
      { product: "Laptop", price: 50000 }
    ]
  },
  {
    customer_id: "C103",
    name: "Neha",
    items: [
      { product: "Earphones", price: 2000 },
      { product: "Laptop", price: 50000 }
    ]
  }
])
< {
    acknowledged: true,
```

SCENARIO 1: Mobile Product Revenue Report

```
> db.orders.aggregate([{$unwind:"$items"},{$match:{"items.product":"Mobile","items.price":{$gte:15000}}},{$group:{_id:"$items.product",
< {
    _id: 'Mobile',
    totalCustomers: 2,
    totalRevenue: 30000,
    avgPrice: 15000,
    minPrice: 15000,
    maxPrice: 15000,
    customerNames: [
      'Asha',
      'Kiran'
    ],
    firstCustomer: 'Asha'
  }
```

SCENARIO 2: High Value Products Report

```
> db.orders.aggregate([{$unwind:"$items"},{$match:{"items.price":{$gte:15000},"items.product":{$in:["Mobile","Laptop"]}}},{$group:{_id:
< {
    _id: 'Mobile',
    orderCount: 2,
    totalRevenue: 30000,
    avgPrice: 15000,
    minPrice: 15000,
    maxPrice: 15000,
    customerList: [
      'Asha',
      'Kiran'
    ]
  }
  {
    _id: 'Laptop',
    orderCount: 2,
    totalRevenue: 100000,
    avgPrice: 50000,
    minPrice: 50000,
    maxPrice: 50000,
    customerList: [
      'Kiran',
      'Neha'
    ]
  }
```

SCENARIO 3: Customer-wise Spending Report

```
> db.orders.aggregate([{$unwind:"$items"},{$group:{_id:"$customer_id",customerName:{$first:"$name"},totalSpent:{$sum:"$items.price"},av
< {
    _id: 'C103',
    customerName: 'Neha',
    totalSpent: 52000,
    avgPrice: 26000,
    minPrice: 2000,
    maxPrice: 50000,
    productsBought: [
      'Earphones',
      'Laptop'
    ],
    productCount: 2
  }
  {
    _id: 'C102',
    customerName: 'Kiran',
    totalSpent: 65000,
    avgPrice: 32500,
    minPrice: 15000,
    maxPrice: 50000,
    productsBought: [
      'Mobile',
      'Laptop'
    ],
    productCount: 2
  }
```

Case Study 2 : Movie Ticket Booking System

```
> db.customers.insertMany([{_id:1,name:"Arun",profile:{age:25,city:"Chennai"}},{_id:2,name:"Priya",profile:{age:22,city:"Bangalore"}},{
< {
    acknowledged: true,
    insertedIds: {
      '0': 1,
      '1': 2,
      '2': 3,
      '3': 4,
      '4': 5
    }
  }
```

```
> db.movies.insertMany([{_id:"M1",title:"Inception",genre:"Sci-Fi",price:300},{_id:"M2",title:"Avatar",genre:"Fantasy",price:350},{_id:
< {
    acknowledged: true,
    insertedIds: {
      '0': 'M1',
      '1': 'M2',
      '2': 'M3'
    }
  }
```

```
> db.bookings.insertMany([{_id:301,customer_id:1,movie_id:"M1",show_date:"2026-02-01"},{_id:302,customer_id:1,movie_id:"M2",show_date:'
< {
    acknowledged: true,
    insertedIds: {
      '0': 301,
      '1': 302,
      '2': 303,
      '3': 304,
      '4': 305,
      '5': 306,
      '6': 307,
      '7': 308
    }
  }
```

Query 1: Total bookings per movie

```
> db.bookings.aggregate([{$group:{_id:"$movie_id",totalBookings:{$sum:1}}}])
< {
    _id: 'M3',
    totalBookings: 2
  }
  {
    _id: 'M1',
    totalBookings: 3
  }
  {
    _id: 'M2',
    totalBookings: 3
  }
```

Query 2: Movie-wise revenue report

```
> db.bookings.aggregate([{$lookup:{from:"movies",localField:"movie_id",foreignField:"_id",as:"movieDetails"}},{$unwind:"$movieDetails"}
< {
    _id: 'M2',
    movieTitle: 'Avatar',
    totalTicketsSold: 3,
    totalRevenue: 1050,
    avgPrice: 350,
    minPrice: 350,
    maxPrice: 350
  }
  {
    _id: 'M1',
    movieTitle: 'Inception',
    totalTicketsSold: 3,
    totalRevenue: 900,
    avgPrice: 300,
    minPrice: 300,
    maxPrice: 300
  }
  {
    _id: 'M3',
    movieTitle: 'Interstellar',
    totalTicketsSold: 2,
    totalRevenue: 800,
    avgPrice: 400,
    minPrice: 400,
    maxPrice: 400
  }
```

Query 3: Customer-wise spending report

```
> db.bookings.aggregate([{$lookup:{from:"movies",localField:"movie_id",foreignField:"_id",as:"movieDetails"}},{$unwind:"$movieDetails"}
< {
    _id: 4,
    moviesBooked: 1,
    totalSpent: 350,
    avgSpending: 350,
    movieList: [
      'Avatar'
    ]
  }
  {
    _id: 3,
    moviesBooked: 2,
    totalSpent: 700,
    avgSpending: 350,
    movieList: [
      'Interstellar',
      'Inception'
    ]
  }
  {
    _id: 1,
    moviesBooked: 2,
    totalSpent: 650,
    avgSpending: 325,
    movieList: [
      'Inception',
      'Avatar'
```

Query 4: Customers with multiple bookings

```
> db.bookings.aggregate([{$group:{_id:"$customer_id",bookingCount:{$sum:1}}},{$match:{bookingCount:{$gt:1}}}])
< {
    _id: 3,
    bookingCount: 2
  }
  {
    _id: 1,
    bookingCount: 2
  }
  {
    _id: 5,
    bookingCount: 2
  }
```

Query 5: Genre-wise booking count

```
> db.bookings.aggregate([{$lookup:{from:"movies",localField:"movie_id",foreignField:"_id",as:"movieDetails"}},{$unwind:"$movieDetails"}
< {
    _id: 'Fantasy',
    totalBookings: 3
  }
  {
    _id: 'Sci-Fi',
    totalBookings: 5
  }
```

Query 6: Movie popularity with customer list

```
> db.bookings.aggregate([{$lookup:{from:"customers",localField:"customer_id",foreignField:"_id",as:"customerDetails"}},{$unwind:"$custo
< {
    _id: 'M2',
    totalBookings: 3,
    customerList: [
      'Arun',
      'Neha',
      'Kiran'
    ]
  }
  {
    _id: 'M1',
    totalBookings: 3,
    customerList: [
      'Arun',
      'Priya',
      'Rahul'
    ]
  }
  {
    _id: 'M3',
    totalBookings: 2,
    customerList: [
      'Rahul',
      'Kiran'
    ]
  }
```